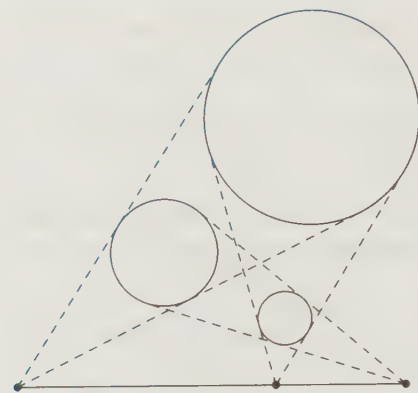




Monge's Theorem

Probably no symbol in mathematics has evoked as much mystery, fascination, and human interest as the number π . – William L. Schaaf

CHAPTER 11

Circles

11.1 Arc Measure, Arc Length, and Circumference

The portion of a circle connecting two points on the circle's circumference is called an **arc**. We find the measure of an arc of a circle by considering what fraction of the circle the arc is. For example, a whole circle is 360° , so an arc that is $1/6$ of the circle has measure $(1/6)(360^\circ) = 60^\circ$.

However, not all 60° arcs are the same. Some are much longer than others, as shown in the figure to the right. Thus, we need more than just angle measure to classify arcs. We need a way to measure the lengths of arcs.



We start with the length of an entire circle, which would be the perimeter of a circle. The perimeter of a circle is called the circle's **circumference**.

Before reading the rest of this chapter, put yourself in the sandals of the ancient Greeks and try a little experiment. Get a string, and find numerous circular objects. Measure the distance around each object by wrapping the string around it. Then measure the diameter of the object. Finally, for each object find the quotient

$$\frac{\text{Distance Around the Object}}{\text{Diameter of Object}}$$

You should find that in each case the quotient is around 3.14. (If you get anything different, try measuring and dividing again!)

In every single circle, the ratio of circumference to diameter is the same. This ratio is called **pi**, and is given the symbol π . Its value is approximately 3.14. (If you ask a bunch of your math friends, you're likely to find someone who knows dozens more digits.)

Sidenote: For some reason, memorizing huge portions of pi is a bit of a sport among math-lovers. This should help you get started:



3.141592653589793238462643383279502884197169399375105820974944
59230781640628620899862803482534211706798214808651328230664709
38446095505822317253594081284811174502841027019385211055596446
22948954930381964428810975665933446128475648233786783165271201
90914564856692346034861045432664821339360726024914127372458700
66063155881748815209209628292540917153643678925903600113305305
48820466521384146951941511609433057270365759591953092186117381
93261179310511854807446237996274956735188575272489122793818301

(Please don't tell your parents you got this π -memorizing idea from us.)

Pi is an **irrational number**, which means that it cannot be expressed as a ratio of integers. Because pi is irrational, its decimal expansion does not terminate and does not become periodic. In other words, it does not get to a point where the same set of numbers is repeated over and over. So, there are no shortcuts to memorizing digits of pi!

Problems

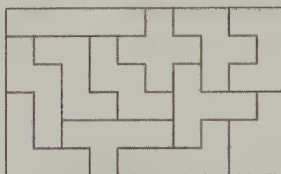
Problem 11.1: The circumference of a circle is 32π . Find the circle's radius.

Problem 11.2: The length of a 72° arc of a circle is 15. What is the circumference of the circle?

Problem 11.3: Chord $\overline{YZ}$ of a circle with center O has length 12. The circumference of the circle is 24π .

- What kind of triangle is $\triangle YOZ$?
- Find the radius of the circle.
- Find the length of $\widehat{YZ}$.

Extra! The twelve pentominoes (see page 132) fit inside a 6×10 rectangle as shown below.



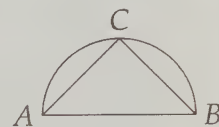
They can also fit inside a 5×12 , a 4×15 , and even a 3×20 rectangle. Can you find how?

Problem 11.4: Chords $\overline{AB}$ and $\overline{CD}$ of $\odot O$ have the same length. We will prove that $\widehat{AB} = \widehat{CD}$.

- Connect A , B , C , and D to the center of the circle. Find some congruent triangles.
- Use your congruent triangles to prove that $\widehat{AB} = \widehat{CD}$.

Problem 11.5:

- Anna and Betty are 180 yards apart, at points A and B in the diagram, respectively. There is a semi-circular path from Anna to Betty. Chuck is at point C , the midpoint of this path. Anna must walk to Chuck and then to Betty. How much farther must Anna walk if she follows the path than if she 'cuts across,' walking straight to C then straight to B ?
 - What if the path is a 120° arc of a circle instead of a semi-circular arc? (Chuck is on the midpoint of this new arc, and we wish to see how much farther Anna must walk by staying on the path.)
- Hints:** 20, 569



Circumference is a pretty simple concept, once you've finished memorizing all the digits of pi.

Problem 11.1: The circumference of a circle is 32π . Find the circle's radius.

Solution for Problem 11.1: Let C be the circumference and d be the diameter. We know that $C/d = \pi$ and $C = 32\pi$. Therefore, $d = 32$. The radius is half the diameter, or $32/2 = 16$. $\square$

Important: In a circle, let C be the circumference, d be the diameter, and r be the radius.



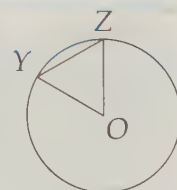
$$C = \pi d = 2\pi r.$$

Problem 11.2: The length of a 72° arc of a circle is 15. What is the circumference of the circle?

Solution for Problem 11.2: A 72° arc is $72^\circ/360^\circ = 1/5$ of an entire circle. Since this arc has length 15, the entire circumference of the circle has length $5(15) = 75$. $\square$

Problem 11.3: Chord $\overline{YZ}$ of a circle with center O has length 12. The circumference of the circle is 24π . Find the length $\widehat{YZ}$.

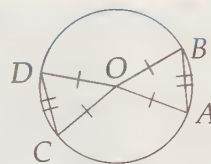
Solution for Problem 11.3: To find the length of the arc, we must find $\angle YOZ$. We don't have any information about angles, however, so we start by figuring out some lengths. Since the circumference is 24π , the diameter is 24 and the radius is 12. Hence, $OY = YZ = ZO$, so $\triangle YOZ$ is equilateral. Therefore, $\angle YOZ$ is 60° , and our arc is $1/6$ of the circle. So, the length of $\widehat{YZ}$ is $(1/6)(24\pi) = 4\pi$. $\square$



A **central angle** of a circle is an angle with the center of the circle as its vertex. Now that we know how to relate arcs to central angles, we can find a useful relationship between the arcs cut off by congruent chords of a circle.

Problem 11.4: Chords $\overline{AB}$ and $\overline{CD}$ of $\odot O$ have the same length. Prove that $\widehat{AB} = \widehat{CD}$.

Solution for Problem 11.4: We form two triangles by connecting A, B, C , and D to O . Since $AB = CD$, $AO = CO$, and $BO = DO$, we have $\triangle ABO \cong \triangle CDO$ by SSS Congruence. Therefore, $\angle AOB = \angle COD$, so arcs $\widehat{AB}$ and $\widehat{CD}$ are congruent. $\square$



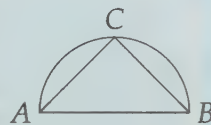
As you might suspect (and will prove as an Exercise), this relationship between congruent chords and congruent arcs works in reverse, too.

Important: If two chords of a circle are congruent, then the arcs they **subtend** (cut off) are congruent. Conversely, if two arcs of a circle are congruent, then the chords that subtend them are congruent.



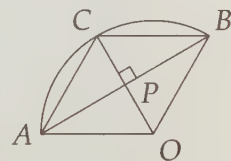
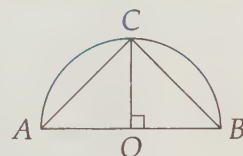
Problem 11.5:

- (a) Anna and Betty are 180 yards apart, at points A and B in the diagram, respectively. There is a semi-circular path from Anna to Betty. Chuck is at point C , the midpoint of this path. Anna must walk to Chuck and then to Betty. How much farther must Anna walk if she follows the path than if she 'cuts across,' walking straight to C then straight to B ?
- (b) What if the path is a 120° arc of a circle instead of a semi-circular arc? (Chuck is on the midpoint of this new arc, and we wish to see how much farther Anna must walk by staying on the path.)



Solution for Problem 11.5:

- (a) We might immediately see that $\triangle ACB$ is a 45-45-90 triangle, but if we don't, we can start by connecting C to the center of the circle, O , as shown in the diagram. Since $\widehat{AC} = \widehat{CB} = 90^\circ$, we have $\angle AOC = \angle COB = 90^\circ$. $\overline{OA}$, $\overline{OB}$, and $\overline{OC}$ are all radii of the circle, so $\triangle AOC$ and $\triangle BOC$ are 45-45-90 triangles. Since $AB = 180$ yards, we have $AO = OB = OC = 180/2 = 90$ yards and $AC = BC = CO\sqrt{2} = 90\sqrt{2}$ yards. Since arc $\widehat{AB}$ is one-half a circle, its length is $(AB)\pi/2 = 90\pi$ yards. So, Anna must walk $90\pi - (AC + BC) = 90\pi - 180\sqrt{2}$ yards farther if she follows the path.
- (b) We start by connecting A, B , and C to the center of the circle of which the path is a part, since we know more about radii than we do about arcs. Since $\widehat{AB}$ is 120° , and C is its midpoint, $\widehat{AC} = \widehat{BC} = 60^\circ$. Therefore, $\angle AOC = \angle COB = 60^\circ$. $OA = OC$ because they are radii of the same circle. Therefore, $\triangle AOC$ is isosceles with $\angle AOC = 60^\circ$ and $AO = OC$, so $\triangle AOC$ is equilateral. Similarly, $\triangle BOC$ is also equilateral.



We need to find the radius of this circle to finish the problem. Let $\overline{AB}$ meet $\overline{CO}$ at P . Since $\triangle AOP \cong \triangle BOP$ by SAS, $\angle APO = \angle BPO$. These angles together make up a straight angle, so $\angle APO = \angle BPO = 90^\circ$. So, $\triangle APO$ and $\triangle BPO$ are 30-60-90 triangles. Since $AP = AB/2 = 90$ yards, we have $AO = AP(2/\sqrt{3}) = 60\sqrt{3}$ yards. Since $AC = AO = BO = BC$, we have $AC + CB = AO + BO = 120\sqrt{3}$. The length of $\widehat{AB}$ is $(120^\circ/360^\circ)(2AO\pi) = 40\pi\sqrt{3}$. Therefore, by following the path, Anna walks $\widehat{AB} - (AC + BC) = 40\pi\sqrt{3} - 120\sqrt{3}$ farther. $\square$

The key step in the second part of this problem is connecting the points on the path to the center of the circle. This is often an important step in challenging geometry problems.

Concept: In complicated problems involving a circle, try connecting the center of a circle to important points on the circle.

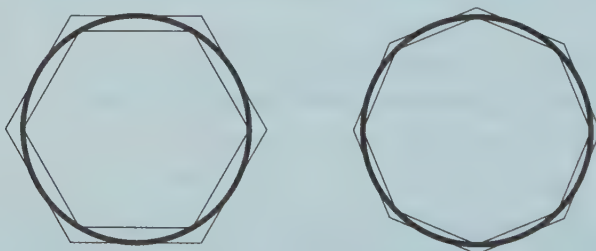
Sidenote: We defined π as the ratio of the circumference of a circle to its diameter. But how do we know what π is numerically? In other words, how do we know that $\pi = 3.1415926535 \dots$, and so on?



Since π is one of the most important mathematical constants, much time and effort have been devoted to calculating π . At the present time, π is known to over one trillion decimal places. Why bother with so many digits? Certainly not for the sake of accuracy in everyday use – even 39 decimal places is sufficient to calculate the circumference of the known universe to within the radius of a hydrogen atom.

The answer, in part, is for the prestige of having set a new record number of places. The digits of π can also be used to check the speed and accuracy of computers. Finally, mathematicians are interested in seeing if they can find any patterns among these digits, such as whether the ten digits 0, 1, 2, 3, $\dots$, 9 appear equally often. However, despite all this effort, no patterns of note have been found.

So how do we calculate the value of π ? One simple method is as follows: Suppose we take a circle of radius 1, and circumscribe it, and inscribe it, with a regular n -gon. The figure below shows the n -gons for $n = 6$ and $n = 8$.



The perimeter of the polygons approximate the circumference of the circle, which is 2π . The greater the number of sides in the polygon, the better the approximation. Let a_n denote the perimeter of the circumscribed n -gon (the outer polygon), and let b_n denote the perimeter of the inscribed n -gon (the inner polygon). Then it turns out that

$$a_{2n} = \frac{2a_n b_n}{a_n + b_n} \quad \text{and} \quad b_{2n} = \sqrt{a_{2n} b_n}.$$

Continued on the next page. . .

Sidenote: ... continued from the previous page



Thus, if we start with a_6 and b_6 , we can calculate a_{12} and b_{12} , which can be used to calculate a_{24} and b_{24} , and so on. This was the observation of the ancient Greek mathematician **Archimedes**, after whom this algorithm is named. The following table shows some results of this algorithm. We begin with the values $a_6 = 4\sqrt{3}$ and $b_6 = 6$. (Make sure you see why $a_6 = 4\sqrt{3}$ and $b_6 = 6$; remember, the radius of the circle is 1.)

n	a_n	b_n	$a_n/2$	$b_n/2$
6	6.92820323	6.00000000	3.46410162	3.00000000
12	6.43078062	6.21165708	3.21539031	3.10582854
24	6.31931988	6.26525723	3.15965994	3.13262861
48	6.29217243	6.27870041	3.14608622	3.13935020
96	6.28542920	6.28206390	3.14271460	3.14103195
192	6.28374610	6.28290494	3.14187305	3.14145247
384	6.28332549	6.28311522	3.14166275	3.14155761
768	6.28322035	6.28316778	3.14161018	3.14158389

As we can see, the sequences $(a_n/2)$ and $(b_n/2)$ both converge to π . This is not the best algorithm for calculating the digits of π , but it illustrates how we can use a simple geometric idea to approximate π .

Exercises

11.1.1 Find the circumference of a circle with radius 4.

11.1.2 Find the radius of a circle with circumference 12π .

11.1.3 $\widehat{AC}$ of $\odot O$ has length 12π , and the circle has radius 18.

(a) Find OC .

(b) Find $\angle AOC$.

(c)★ Find AC . **Hints:** 471

11.1.4 Arcs $\widehat{WX}$ and $\widehat{YZ}$ of $\odot Q$ are congruent. Prove that $WX = YZ$.

Extra! Degrees are not the only way to measure angles! Just as there are different units for measuring length, like inch and centimeter, there are different units for angle measures. Besides the degree, the most commonly used unit for angle measure is the **radian**. Just as there are 360 degrees in a circle, there are 2π radians in a circle. Therefore, the measure in radians of a quarter-circle is $(2\pi)/4 = \pi/2$ radians, and of a semicircle is $(2\pi)/2 = \pi$ radians.

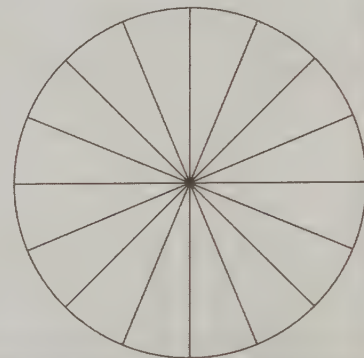
As you might guess, the usage of radians is inspired by circumference. You might also notice that every arc of a circle with radius 1 has its length in units equal to its measure in radians. However, radians are not very useful until our study of advanced trigonometry and calculus, so we'll stick with degrees for now.

11.2 Area

Problems

Problem 11.6: In this problem we will develop a formula for the area of a circle with radius r .

- The circle in the diagram has radius r and is divided into 16 equal pieces by drawing 8 diameters. These pieces are called **sectors**. Rearrange the sectors to look very similar to a quadrilateral whose area we can find.
- What is the length of the base of this 'quadrilateral' in terms of r ?
- What is the height of this 'quadrilateral' in terms of r ?
- What is the area of this 'quadrilateral' in terms of r ?



Problem 11.7: Find the area of a circle with diameter 12.

Problem 11.8: $\triangle XYZ$ is equilateral with side length 10. A circle is constructed with center X and radius 10, thus passing through Y and Z . Find the area of sector YXZ of the circle. (In other words, find the area of the portion of the circle bounded by radii $\overline{XY}$ and $\overline{XZ}$ and by arc $\widehat{YZ}$.)

Problem 11.9: Farmer Tim has 50 feet of fence. He wants to enclose a semicircular area adjacent to his barn, thus using his barn as one side of the enclosure. What is the area of the space Farmer Tim can enclose?

Having tackled the 'perimeter' of a circle, we now turn to its area. While proving the formula for the area of a circle would require far more advanced tools than we have now, we can develop an intuitive explanation for the formula.

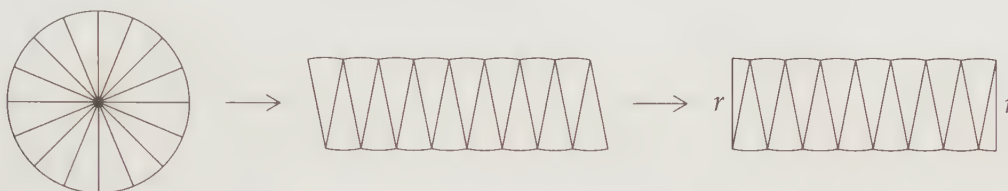
Problem 11.6: Find a formula for the area of a circle by cutting up a circle and rearranging the pieces to look a lot like a rectangle.

Extra! GggXyZ the Martian has been told by the Martian King to make a rope to lasso the Earth.

➡➡➡➡➡ GggXyZ makes a rope that will fit snugly around the equator of the Earth. The Martian King decides he wants a little slack so he can put the rope around the Earth more easily. He wants the rope to fit around the equator such that there is one inch between the Earth and the rope all the way around the equator.

GggXyZ says this is no problem. She pulls out a short piece of rope and says she can just add this little piece to her rope to make the rope long enough. The Martian King laughs at her, saying she'll need miles of rope because the diameter of the Earth is almost 8,000 miles. Who is right? How much rope does she need?

Solution for Problem 11.6: We start by cutting the circle into equal slivers, which we call **sectors**, by drawing equally spaced radii as shown below. We then rearrange the sectors as shown in the second figure, thus forming a figure that resembles a parallelogram. Finally, we can take half of one of the end sectors and slide it to the other end as shown in the final diagram. Now we have a figure that strongly resembles a rectangle.



The 'width' of this 'rectangle' is clearly the radius of the circle, which we'll call r . The 'length' of the 'rectangle' is half the circumference of the circle, since the circumference of the circle is equally divided among the top and bottom of our 'rectangle.' Hence, the 'length' is $(2\pi r)/2 = \pi r$. Since the area of a rectangle is its length times its width, the area of our 'rectangle' is $(\pi r)(r) = \pi r^2$.

This is not a rigorous proof, but it should be clear that if we cut the circle into more and more sectors and do the described rearrangement, the 'rectangle' looks more and more like a real rectangle. $\square$

We now know how to find the area of a circle.

Important: The area of a circle with radius r is πr^2 .



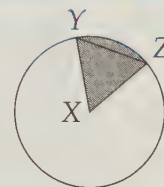
Let's use our newfound knowledge of how to find the area of a circle on a few problems.

Problem 11.7: Find the area of a circle with diameter 12.

Solution for Problem 11.7: Since the diameter is 12, the radius is 6. Therefore, the area is $(6^2)\pi = 36\pi$. $\square$

Problem 11.8: $\triangle XYZ$ is equilateral with side length 10. A circle is constructed with center X and radius 10, thus passing through Y and Z . Find the area of sector YXZ of the circle.

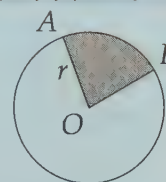
Solution for Problem 11.8: Our sector is shaded in the diagram. To find its area, we must determine what portion the sector is of the whole circle. Since $\triangle XYZ$ is equilateral, $\angle YXZ = 60^\circ$. Since the whole circle is 360° , this means that our sector is $60^\circ/360^\circ = 1/6$ of the entire circle. The whole circle has area $\pi r^2 = \pi(10^2) = 100\pi$, so the sector has area $(100\pi)(1/6) = 50\pi/3$. $\square$



Important: Following our solution to Problem 11.8, we can find the area of a sector of a circle of radius r given the central angle of the sector. In the figure to the right, $\angle AOB$ is $(\angle AOB)/360^\circ$ of the whole circle. Therefore, we have:



$$\text{Area of sector } AOB = \left(\frac{\angle AOB}{360^\circ} \right) r^2 \pi.$$



You should not have to memorize this formula. Instead, understand that the area of a sector is simply related to the ratio of the sector's angle to the number of degrees in a whole circle.

Problem 11.9: Farmer Tim has 50 feet of fence. He wants to enclose a semicircular area adjacent to his barn, using his barn as one side of the enclosure. What is the area of the space Farmer Tim can enclose?

Solution for Problem 11.9: We need to find the radius of the semicircle in order to find its area. We start with a sketch of the problem. Our fence forms a semicircular arc, so the arc has length 50. Therefore, the semicircle is half of a circle that has circumference 100. The diameter of such a circle is the circumference divided by π , or $100/\pi$. So, the radius of our semicircle is $(100/\pi)/2 = (50/\pi)$.



The semicircle therefore has an area that is $1/2$ the area of a circle with radius $(50/\pi)$, or $(1/2)(50/\pi)^2(\pi) = 1250/\pi$ square feet. $\square$

Concept: Start geometric word problems with a sketch.



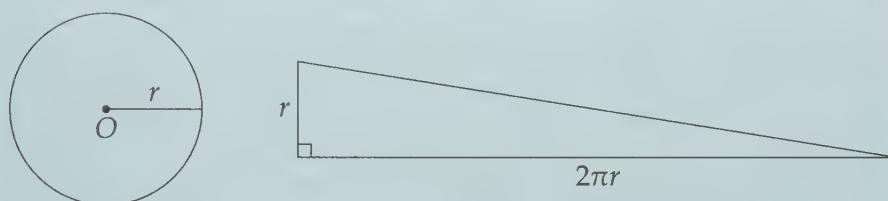
Exercises

- 11.2.1** Find the area of a circle with diameter 18.
- 11.2.2** Find the area of a circle that has circumference 12π .
- 11.2.3** Find the radius of a circle if its circumference is numerically equal to its area.
- 11.2.4** If a pizza that is 12 inches in diameter provides four full meals, how many meals are provided by a pizza that is 20 inches in diameter?
- 11.2.5** Points A and B are on the circumference of $\odot O$ such that $\angle AOB = 120^\circ$ and $OA = 12$.
- (a) Find the area of $\odot O$.
- (b) Find the area of sector AOB .
- 11.2.6** Sector XQZ of circle Q has area 30π . Given that the whole circle has area 100π , find $\angle XQZ$ and $\angle XZQ$. **Hints:** 342
- 11.2.7★** A man standing on a lawn is wearing a circular sombrero of radius 3 feet. Unfortunately, the hat blocks the sunlight so effectively that the grass directly under it dies instantly. If the man walks in a circle of radius 5 feet, what area of dead grass will result? (Source: HMMT) **Hints:** 81

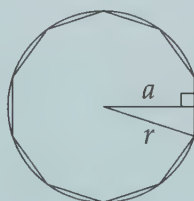
Extra! In 1897, the Indiana state legislature almost passed a bill that set the value of π to exactly 3.2. The House voted unanimously for it and it passed a first reading in the Senate. Fortunately, a math professor at Purdue University happened to be visiting the legislature at the same time and advised that the bill be postponed indefinitely, effectively killing it. If he hadn't, manholes in Indiana would look very strange.

Sidenote: The ancient Greeks didn't have calculus, so how did they find the formula for the area of a circle? While they didn't have calculus, they did have **Archimedes**, some of whose proofs came so close to calculus (which wouldn't be developed until nearly 2000 years after his death) that some suggest he might be called the father of calculus. Archimedes used two favorite tactics of the Greeks – contradiction (see page 41) and comparing a desired area to that of a figure whose area is easy to find.

Archimedes claimed that a circle has the same area as that of a right triangle with the circle's radius as one leg and its circumference as the other leg. He did so by proving that the area of the circle could be neither more nor less than that of the triangle. We'll show you his proof that the circle's area could not be greater than that of the triangle, and leave you to walk in his footsteps and supply the proof for the other half.



Let the radius of the circle be r , the area of the circle be A , and the area of the triangle be T . Then, we assume that $A > T$. As we saw in our discussion of calculating π , we can use regular polygons with more and more sides to approximate a circle. Specifically, at some point, the difference between the area of such a polygon, which we'll call P , and the area of the circle is less than $A - T$. Therefore, we have $A - P < A - T$, so $T < P$.



However, we showed on page 252 that the area of such a regular polygon is one-half the product of its perimeter, p , and its apothem, a . Since the perimeter of the polygon is smaller than the perimeter of the circle, we have $p < 2\pi r$. Moreover, the apothem of the polygon is clearly less than the radius of the circle. Therefore, we can use $p < 2\pi r$ and $a < r$ to write $P = ap/2 < (r)(2\pi r)/2 = T$. So, we have both $T < P$ and $P < T$. This is impossible, so we have reached a contradiction. We can thus conclude that our assumption $A < T$ was false.

See if you can complete the proof by showing that it is impossible to have $A > T$. As you might guess, you'll have to consider a polygon circumscribed about the circle!

Source: *Journey Through Genius* by William Dunham

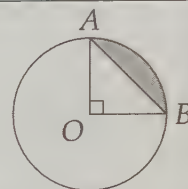
11.3 Funky Areas

We now know how to tackle the area of sectors and of triangles. We can put these two basic tools together to find the areas of all sorts of funky figures.

Problems

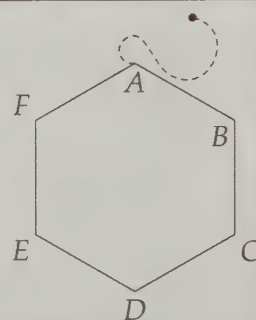
Problem 11.10: O is the center of the shown circle and $OA = 8$. The shaded region between chord $\overline{AB}$ and the circle is called a **circular segment**.

- Find the area of sector AOB .
- Find the area of $\triangle AOB$.
- Find the area of the shaded region.



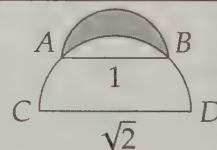
Problem 11.11: I have a barn that is a regular hexagon, as shown. Each side of the barn is 100 feet long. I tether my burro to point A with a 150 foot rope. In this problem we will find the area of the region in which my burro can graze.

- Imagine you are the burro. Sketch out the area in which you can graze. Pay close attention to what happens when you try to go past point B towards point C .
- Break your grazing region into sectors whose areas you can find, then find the area of the grazing region.



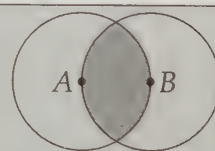
Problem 11.12: The shaded portion of the figure is called a **lune**. Given that $AB = 1$, $CD = \sqrt{2}$, and that $\overline{AB}$ and $\overline{CD}$ are diameters of the respective semicircles shown, we wish to find the area of the lune. (*Source: AMC 10*)

- Let O be the center of the larger semicircle. Draw $\overline{OA}$ and $\overline{OB}$. How long are these segments? What kind of triangle is $\triangle AOB$?
- Find the area of circular segment AB (i.e., the unshaded part of the small semicircle).
- Find the area of the lune.



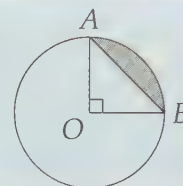
Problem 11.13: Circle A passes through B and circle B passes through A . Given that $AB = 6$, we wish to find the area of the shaded region common to both circles.

- Let the two points where the circles intersect be X and Y . Connect A and B to each other and to X and Y .
- What kind of triangles are $\triangle XAB$ and $\triangle YAB$?
- We know how to find the areas of circles, triangles, and circular segments. Find an expression for the shaded area in terms of pieces of the diagram whose areas you can find.
- Find the shaded area.



We know how to find the areas of circles, sectors of circles, triangles, and lots of other polygons. In this section we'll use that knowledge to find the area of even funkier shapes.

Problem 11.10: O is the center of the shown circle and $OA = 8$. The shaded region between chord $\overline{AB}$ and the circle is called a **circular segment**. Find the area of this circular segment.



Solution for Problem 11.10: We view the circular segment in terms of pieces whose areas we can find. The circular segment is what's left when we cut $\triangle AOB$ out of sector AOB . These areas are easy to find, so we have

$$\begin{aligned}\text{Area of circular segment } AB &= \text{Area of sector } AOB - \text{Area of } \triangle AOB \\ &= \left(\frac{90^\circ}{360^\circ}\right) \pi(8^2) - \frac{(AO)(BO)}{2} \\ &= 16\pi - 32.\end{aligned}$$

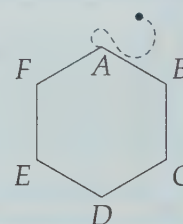
□

Concept:  Nearly all funky area problems are solved by expressing the funky area as sums and/or differences of pieces whose areas we can easily find. The first step should be clearly expressing the funky area in terms of simple areas, as we did in Problem 11.10 when we wrote

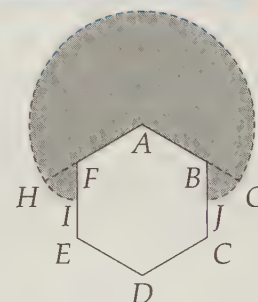
$$\text{Area of circular segment } AB = \text{Area of sector } AOB - \text{Area of } \triangle AOB.$$

Sometimes our region isn't already drawn for us.

Problem 11.11: I have a barn that is a regular hexagon, as shown. Each side of the barn is 100 feet long. I tether my burro to point A with a 150 foot rope. Find the area of the region in which my burro can graze.



Solution for Problem 11.11: First we have to figure out what the burro's region looks like. Say the burro is tethered at A and roams so far that the rope pulls taut. Suppose the burro keeps the rope taut and walks until part of the rope coincides with $\overline{AB}$. The burro then is at point G on $\overline{AB}$. Since $AB = 100$ feet and the rope is only 150 feet long, when the burro moves beyond G towards C , there's only 50 feet of rope for the burro to use. Hence, once the burro moves past G , the region he can reach is just sector JBG . We have the same situation on the other side, leading to sector HFI . Thus, our desired area is



$$\text{Area of major sector } HAG + \text{Area of sector } JBG + \text{Area of sector } HFI.$$

Since $\angle ABC = 120^\circ$ (because $ABCDEF$ is a regular hexagon), we have $\angle JBG = 180^\circ - 120^\circ = 60^\circ$. Therefore, sector JBG is $60^\circ/360^\circ = 1/6$ of a circle with radius 50 feet. Hence, its area is $(1/6)\pi(50^2) = 1250\pi/3$ square feet.

On the other end of the burro's range is sector HFI , which is exactly the same as sector JBG , so it contributes $1250\pi/3$ ft² to the burro's area. Finally we have the sector between major arc $\widehat{GH}$ and radii $\overline{AG}$ and $\overline{AH}$. Since $\angle FAB = 120^\circ$, the sector is $(360^\circ - 120^\circ)/360^\circ = 2/3$ of a circle. The radius of this sector is 150 feet, so its area is $(2/3)\pi(150^2) = 15000\pi$.

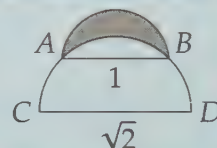
Combining these three parts gives us a grazing area of

$$15000\pi + 2\left(\frac{1250\pi}{3}\right) = \frac{47500\pi}{3} \text{ square feet.}$$

□

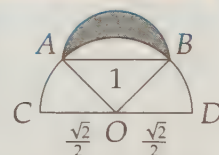
Once again, notice that one of our key steps is to express our desired region in terms of not-so-funky pieces we know how to tackle.

Problem 11.12: The shaded portion of the figure is called a **lune**. Given that $AB = 1$, $CD = \sqrt{2}$, and that $\overline{AB}$ and $\overline{CD}$ are diameters of the respective semicircles shown, find the area of the lune. (Source: AMC 10)



Solution for Problem 11.12: We start by noting that if we can find the area of circular segment AB , then we can find the area of the lune:

$$\text{Area of lune} = \text{Area of small semicircle} - \text{Area of circular segment } AB.$$



To get the area of the circular segment, we start by connecting its endpoints to the center, O , of its circle. Since

$$\text{Area of circular segment } AB = \text{Area of sector } AOB - \text{Area of } \triangle AOB,$$

we have

$$\begin{aligned} \text{Area of lune} &= \text{Area of small semicircle} - \text{Area of circular segment } AB \\ &= \text{Area of small semicircle} - (\text{Area of sector } AOB - \text{Area of } \triangle AOB) \\ &= \text{Area of small semicircle} - \text{Area of sector } AOB + \text{Area of } \triangle AOB. \end{aligned}$$

To find the area of sector AOB , we need $\angle AOB$. Since $CD = \sqrt{2}$, we have $OC = OA = OB = OD = \sqrt{2}/2$. Seeing those $\sqrt{2}$ s makes us think of 45-45-90 triangles. We check and see that indeed $AO^2 + BO^2 = 1/2 + 1/2 = 1 = AB^2$, so $\angle AOB$ is a right angle since the sides of $\triangle AOB$ satisfy the Pythagorean Theorem.

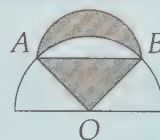
We can now find the areas of all of our pieces. The radius of the small semicircle is $1/2$, so its area is $\pi(1/2)^2/2 = \pi/8$. Sector AOB is $90^\circ/360^\circ = 1/4$ of a circle with radius $\sqrt{2}/2$, so its area is $(1/4)\pi(\sqrt{2}/2)^2 = \pi/8$. Finally, $[AOB] = (AO)(OB)/2 = 1/4$, so we have

$$\text{Area of lune} = \frac{\pi}{8} - \frac{\pi}{8} + \frac{1}{4} = \frac{1}{4}.$$

□

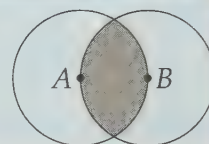
Again, look at our key step – expressing our funky area (the lune) in terms of not-so-funky areas (semicircle, sector, triangle).

Sidenote: You might have noticed that the area of the lune in Problem 11.12 equals the area of $\triangle AOB$. This isn't a coincidence! Moreover, the discovery that the area of the lune described in Problem 11.12 equals the area of $\triangle AOB$ led to one of the earliest notable proofs in math history.

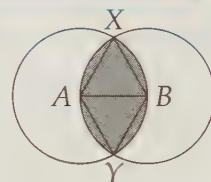


As we'll discuss on page 338, mathematicians tried in vain to square the circle (to 'square the circle' means construct a square with the same area as a given circle). However, the Greek mathematician Hippocrates succeeded in squaring the lune using the insight that the area of the lune of Problem 11.12 equals the area of the triangle with the lune's diameter and the center of the larger semicircle as vertices. See if you can walk in his footsteps by both re-creating the proof, and by performing the construction.
Source: *Journey Through Genius* by William Dunham

Problem 11.13: Circle A passes through B and circle B passes through A. Given that $AB = 6$, find the area of the shaded region common to both circles.



Solution for Problem 11.13: We have a funky circular area, so we start by drawing some segments to cut the region into pieces we can handle. Some obvious candidates are $\overline{AB}$ connecting the centers, and the segments from the centers to the points where the circles meet. We thus form some triangles, sectors, and circular segments. We know how to handle these – we start by finding angles.



Since each of our segments is a radius of one (or both) of the circles, all of them are equal in length. Specifically, $AX = XB = AB = AY = YB = 6$. Therefore our two triangles are equilateral, so all of their angles are equal. We're now ready to express the shaded region in terms of pieces we can handle. We can do so in a few different ways. Here's one way:

$$\text{Shaded Area} = \text{Area of sector } XBY + \text{Area of sector } XAY - \text{Area of } XAYB.$$

We subtract $[XAYB]$ because when we add the sectors, we include the overlap, $[XAYB]$ twice. We could also have written

$$\text{Shaded Area} = [XAYB] + 4(\text{Area of circular segment } XB),$$

since all the circular segments are congruent. We'll take the first approach here since sectors are easier to deal with than circular segments. Since $\angle XBY = \angle XBA + \angle YBA = 120^\circ$, the area of sector XBY is $(120^\circ/360^\circ)\pi(6^2) = 12\pi$. Similarly, the area of sector XAY is 12π . Finally, $XAYB$ consists of two equilateral triangles with side length 6. Each has area $6^2 \sqrt{3}/4 = 9\sqrt{3}$, so $[XAYB] = 2(9\sqrt{3}) = 18\sqrt{3}$. Finally, we have

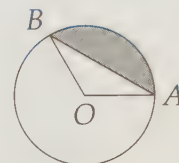
$$\text{Shaded Area} = 24\pi - 18\sqrt{3}.$$

□

Concept: In problems involving multiple circles, connecting the centers can be helpful. In problems involving intersecting circles, connecting the intersection points to the centers (and to each other) is often useful.

Exercises

11.3.1 Find the area of the shaded region given that O is the center of the circle, $\angle AOB = 120^\circ$, and the radius of the circle is 6. **Hints:** 78



11.3.2 Each side of equilateral $\triangle XYZ$ has length 9. Find the area of the region inside the circumcircle of the triangle, but outside the triangle.

11.3.3 In the diagram at left below, $\triangle ABC$ is an equilateral triangle with side length 6. Arcs are drawn centered at the vertices connecting midpoints of consecutive sides, as shown. Find the area of the shaded region.

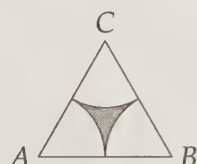


Figure 11.1: Diagram for Problem 11.3.3

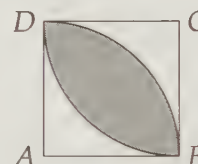


Figure 11.2: Diagram for Problem 11.3.4

11.3.4 In the diagram at right above, quarter-circles have been drawn centered at vertices A and C of square $ABCD$. Given that $AB = 6$, find the shaded area.

11.3.5★ In the diagram at left below, XOY is a quarter-circle. Semicircles are drawn with diameters $\overline{OX}$ and $\overline{OY}$ as shown. Find the area of the shaded region given that $XO = 4$. **Hints:** 405, 458

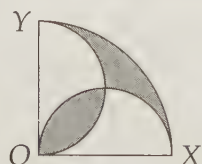


Figure 11.3: Diagram for Problem 11.3.5

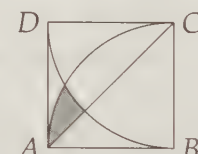


Figure 11.4: Diagram for Problem 11.3.6

11.3.6★ In the diagram at right above, $ABCD$ is a square of side length 4. Two quarter-circles and a diagonal are drawn as shown. Find the area of the shaded region. **Hints:** 121, 485, 450

Extra! $22/7$ is a common approximation of π . $355/113$ is an even better approximation. In 1914, the great Indian mathematician **Ramanujan** provided the uncanny approximation

$$\sqrt[4]{9^2 + \frac{19^2}{22}}.$$

How close is this to π ? (And how in the world did he find this?)

11.4 Summary

Definitions:

- The set of all points that are the same distance from a given point is a **circle**. The given point is the **center** of the circle, and the fixed distance is the **radius**.
- We often refer to a circle by its center using the symbol $\odot$: $\odot O$ refers to a circle with center O .
- A line that touches a circle in a single point is **tangent** to the circle, while a line that hits a circle in two points is a **secant**. A segment connecting two points on a circle is a **chord**, and a chord that passes through the center of its circle is a **diameter**.
- The portion of a circle that connects two points on the circle is an **arc**, which we denote with the endpoints of the arc: $\widehat{MN}$ is the shorter arc that connects M and N .
- The perimeter of a circle is called the circle's **circumference**.

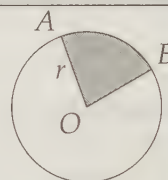
Important: If a circle has diameter d and radius r , then:



- $d = 2r$.
- The circumference of the circle is πd , or $2\pi r$.
- The area of the circle is πr^2 .

Definitions: A portion of a circle cut out by drawing two radii of the circle is called a **sector** of the circle. A portion of a circle between a chord and the arc of the circle connecting the endpoints of the chord is a **circular segment** of the circle.

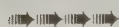
Important: We can find the area of a sector of a circle of radius r given the central angle of the sector. In the figure to the right, since $\angle AOB$ is $(\angle AOB)/360^\circ$ of the whole circle, we have:



$$\text{Area of sector } AOB = \left(\frac{\angle AOB}{360^\circ} \right) r^2 \pi.$$

We can find the area of a circular segment by first finding the area of the sector formed by drawing radii to the endpoints of the segment's chord, then subtracting the area of the triangle with these radii and the chord as sides.

Extra!



$$\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2} + \frac{1}{8^2} + \cdots$$

Problem Solving Strategies

Concepts:

- In complicated problems involving a circle, try connecting the center of the circle to important points on the circle.
- Start geometric word problems with a sketch.
- Nearly all funky area problems are solved by expressing the funky area as sums and/or differences of pieces whose areas we can easily find. The first step should be clearly expressing the funky area in terms of simple areas.
- In problems involving multiple circles, connecting the centers can be helpful. In problems involving intersecting circles, connecting the intersection points to the centers (and to each other) is often useful.

REVIEW PROBLEMS



11.14 Circle O has radius $3\sqrt{3}$. Points A and Z are on the circumference of the circle such that $\angle AOZ = 90^\circ$.

- Find the area of $\odot O$.
- Find the circumference of $\odot O$.
- Find the length of $\widehat{AZ}$.
- Find the area of sector AOZ .
- Find the area of circular segment AZ .

11.15 A giant earth-mover has rubber circular tires 11.5 feet in diameter. Given that there are 5280 feet in a mile, how many revolutions does each tire make during a 6-mile trip? (Answer to the nearest full revolution.) (Source: MATHCOUNTS)

11.16 Find the length of a 78° arc of a circle that has radius 14.

11.17 Find a formula that expresses the area, A , of a circle in terms of its circumference, C .

11.18 Radius $\overline{OA}$ of $\odot O$ is a diameter of $\odot B$. Radius $\overline{OB}$ of $\odot B$ is a diameter of $\odot C$.

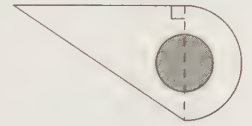
- Find the ratio of the area of $\odot C$ to the area $\odot O$.
- Find CA/OA .

11.19 A 36° arc of $\odot C$ is 24π units long. Find the circumference and the area of the circle.

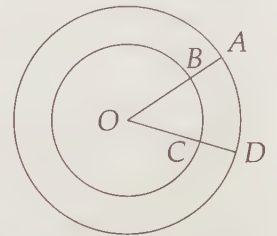
11.20 Regular hexagon $ABCDEF$ is inscribed in $\odot O$ with radius 6. What is the ratio of the circumference of the circle to the perimeter of the hexagon? **Hints:** 106

11.21 On each side of a right triangle a semicircle is constructed using that side as a diameter. How many square centimeters are in the area of the semicircle on the hypotenuse of a right triangle if the areas of the semicircles on the legs of the triangle are 36 and 64 square centimeters? (Source: MATHCOUNTS)

11.22 What is the number of square centimeters in the area that is not shaded in the diagram at right? The radius of the large semicircle is 1 centimeter, the radius of the small circle is 0.5 centimeters, and the length of the longer leg on the right triangle is 3 centimeters. (Source: MATHCOUNTS)



11.23 The larger circle at right has radius 1.5 times the smaller circle. Compute the ratio of the partial ring $ABCD$ to the area of sector BOC . (Source: ARML)

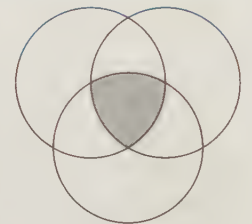


11.24 A rectangle that is 12 inches long and 5 inches wide is inscribed in a circle. What is the area of the region that is inside the circle but outside the rectangle? (Source: MATHCOUNTS)

11.25 Charlyn walks completely around the boundary of a square whose sides are exactly 5 km long. From any point on her path she can see exactly 1 km horizontally in all directions. What is the area of the region consisting of all points Charlyn can see during her walk? (Source: AMC 10)

Challenge Problems

11.26 Three circles of radius 12 lie in a plane such that each passes through the center of the other two. Find the area common to all three circles. **Hints:** 479



11.27 When my car has wheels with a diameter of 24 inches, its speedometer reports the correct speed of my car. I recently replaced my 24-inch wheels with 28-inch diameter wheels. I didn't change my speedometer, however. When the speedometer tells me the car is going 40 miles per hour (and I'm driving with my 28-inch wheels), how fast is my car really going? **Hints:** 515

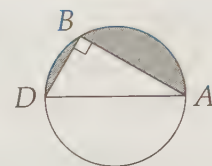
11.28 Two congruent circular coins, $\mathcal{A}$ and $\mathcal{Z}$ are touching at point P . $\mathcal{A}$ is held stationary while $\mathcal{Z}$ is rolled around it one time in such a way that the two coins remain tangent at all times. How many times will $\mathcal{Z}$ revolve around its center? (Source: MATHCOUNTS) **Hints:** 568

11.29 The number of centimeters in the perimeter of a semicircle is numerically the same as the number of square centimeters in its area. What is the number of centimeters in the radius of the semicircle? (Source: MATHCOUNTS) **Hints:** 457

11.30 A cross-section of a river is a trapezoid with bases 10 ft and 16 ft and slanted sides of length 5 ft. At this section the water is flowing at π mph. A little ways downstream is a dam where the water flows through 4 identical circular holes at 16 mph. What is the radius of the holes? **Hints:** 383

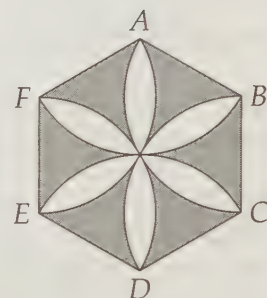
Extra! March 14 (3/14) is known as Pi day.

11.31 Given that $\triangle ABD$ in the diagram at right is a right triangle with $BD = 8$ and $AB = 8\sqrt{3}$, find the total area of the shaded regions. **Hints:** 559



11.32 A circle of radius 3 passes through the center of a square with side length 2. Find the positive difference between the areas of the nonoverlapping portions of the figures. (Source: HMMT) **Hints:** 126

11.33 Six 120° arcs are drawn centered at the vertices of regular hexagon $ABCDEF$. Each arc connects two vertices of the hexagon as shown in the diagram at right. Given that $AF = 6$, find the total area of the shaded regions. **Hints:** 114



11.34 The number of inches in the perimeter of an equilateral triangle equals the number of square inches in the area of its circumscribed circle. What is the radius of the circle? (Source: AMC 10)

11.35 A circle has two parallel chords of length x that are x units apart. If the part of the circle included between the chords has area $2 + \pi$, find x . (Source: HMMT)

11.36★ Quarter circles are drawn centered at each vertex of square $ABCD$ as shown at left below. Given that $AB = 12$, find the area of the shaded region. **Hints:** 217, 272, 51

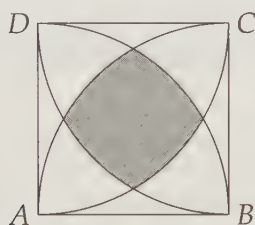


Figure 11.5: Diagram for Problem 11.36

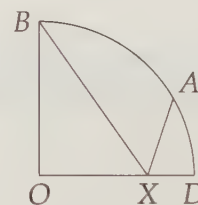


Figure 11.6: Diagram for Problem 11.37

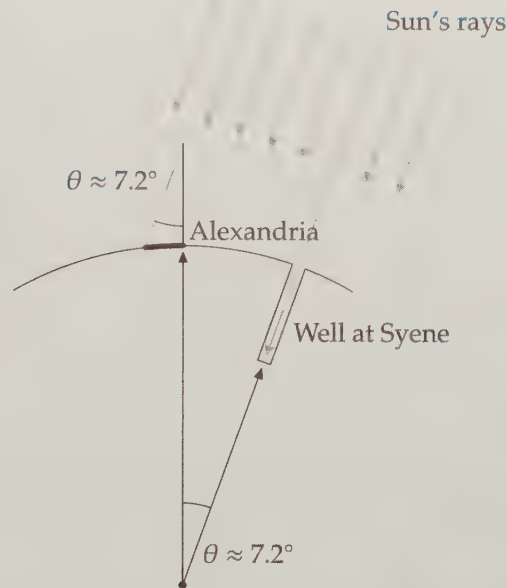
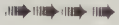
11.37★ The figure at right above shows a quarter-circle of radius 1, with A on $\widehat{BD}$ such that $\angle AOD = 30^\circ$. What must the distance OX be such that the region bounded by $\overline{AX}$, $\overline{BX}$, and $\widehat{AB}$ occupies half the area of the quarter circle? **Hints:** 514, 174, 304

Extra! In the third century B.C., the Greek scholar **Eratosthenes** (the same man who invented the prime number sieve) was the director of the great library at Alexandria. At the time, other Greek scholars knew that the Earth was round, but had no idea how large it was. Eratosthenes learned of a deep well in Syene, a city south of Alexandria, whose bottom was sunlit at noon every year on the summer solstice. This meant that the sun was directly overhead. This gave Eratosthenes an ingenious idea: He could use this observation to measure the size of the Earth.

He assumed that since the sun was so far away, the rays coming from the sun were virtually parallel. Eratosthenes then measured a shadow (some say of a stick, some say of a tower) in Alexandria on the summer solstice, and found that the rays made an angle of approximately 7.2° with the object. Eratosthenes then required one more piece of data: the distance between Alexandria and Syene, which he estimated at 800 km.

Continued on the next page. . .

Extra! . . . continued from the previous page



Since there are 360° in a circle, Eratosthenes estimated the circumference of the Earth to be

$$800 \text{ km} \times \frac{360^\circ}{7.2^\circ} = 40,000 \text{ km.}$$

This is remarkably close to the actual figure, which is approximately 40,075.16 km.

Eratosthenes used an ancient measure of distance called a “stadium,” whose actual length in today’s units is uncertain. Also, Syene is 729 km, not 800 km, from Alexandria and slightly southwest instead of due south. Furthermore, Syene is not exactly on the Tropic of Cancer (where the sun is directly overhead on the summer solstice), but 55 km north. So a number of errors seem to have fortuitously cancelled out to give Eratosthenes such an accurate figure.

What is remarkable about this experiment is not the accuracy of Eratosthenes’ estimate, but rather the conception of the experiment itself. As Dave Hanes, professor of astronomy at Queen’s University, puts it on his website,

The critical point is that Eratosthenes recognized the nature of the problem, found a method, and was able to derive an answer that was correct in spirit in the sense that **he correctly deduced that the Earth was an immense body that was very much larger in extent than the then-known lands of the Mediterranean basin, the home of Greek civilization at the time.** The sense of the discovery is the wonderful thing, not the mere accident that the numerical value was also correct.

Even if Eratosthenes had been off by a factor of two or more in his calculations, it would not have taken away from his brilliant insight into taking a simple observation of the sunlit well and deducing the circumference of the Earth, demonstrating the power of a simple idea.